Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	31 January 2025
Team ID	HDI Predictor Team
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Preparation	USN-1	Load the raw HDI data and inspect the available columns.	2	High	HDI Predictor Team
Sprint-1	Data Cleaning	USN-2	Handle missing values and save the cleaned dataset.	3	High	HDI Predictor Team
Sprint-1	Exploratory Data Analysis	USN-3	Visualize the HDI distribution and feature relationships.	3	Medium	HDI Predictor Team
Sprint-2	Feature Engineering	USN-4	Select the final features and split the dataset.	2	High	HDI Predictor Team
Sprint-2	Model Training	USN-5	Train the Linear Regression model on the selected features.	3	High	HDI Predictor Team
Sprint-3	Model Evaluation	USN-6	Evaluate the model and save it as HDI.pkl.	4	High	HDI Predictor Team
Sprint-3	Flask Integration	USN-7	Build Flask routes and connect the saved model.	4	High	HDI Predictor Team
Sprint-4	Frontend and Testing	USN-8	Develop the UI, test the workflow, and prepare the final demo.	4	High	HDI Predictor Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	4 Days	Day 1	Day 4	8	Day 4
Sprint-2	5	4 Days	Day 5	Day 8	5	Day 8
Sprint-3	8	4 Days	Day 9	Day 12	8	Day 12
Sprint-4	4	3 Days	Day 13	Day 15	4	Day 15

Velocity:
The HDI Predictor project divides work across data preparation, EDA, modeling, Flask integration, testing, and demo delivery so each phase can be tracked clearly.

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

The burn down chart would show the amount of work decreasing as preprocessing, modeling, integration, and testing are completed.

Reference: HDI Predictor project context from the SmartBridge handouts.

Sprint 1 focuses on data loading and cleaning.

Sprint 2 covers exploratory data analysis and feature engineering.

Sprint 3 covers model training, evaluation, and serialization.

Sprint 4 covers Flask integration, frontend development, and testing.

The schedule reflects the 15-day project flow from raw data to the final browser demo.

The backlog is organized so each phase can be reviewed independently.

The project can be demonstrated from notebook to browser in a linear workflow.

The final delivery combines data, model, interface, and validation results.